

Programming and Data Structures Laboratory (CS19003)

Assignment 6 (Section 6): Pointers & String

Department of CSE, IIT Kharagpur

18th March, 2026

General instructions to be followed strictly:

- Name your file as `<roll_no>_<assignment_no>_<question_no>`. For example, if your roll number is 14CS10001 and you are submitting assignment 3 question 1, then name your file as `14CS10001_3_1.c`
 - Write your name, roll number, and assignment number at the beginning of your program.
 - Use proper indentation in your code and comment. Marks may be deducted without proper indentation and comments.
 - Submitting the **correct file** is your responsibility, any request of re-submission after the lab will **never** be entertained.
 - **Zero tolerance** policy will be strictly followed for **plagiarism**. The marks will be put straight zero if such cases are identified.
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Question-1 (15 points)

Write a C program that asks an user to enter an integer from `main()`. Assume that the integer can be at max 4 digit number and can be positive or negative. Write a function `'intToString()'` that converts the integer to string with each character storing one digit of the integer (as the corresponding character). If the integer is negative, the leftmost (first) character in the string representation of the number has to be '-' (the minus sign). However, for positive integers, the string representation should not have explicit '+' sign in front.

`'intToString()'` should have two input parameters, 1. The integer to convert to string 2. The string (character array) which will store the string representation. Call this function from `main()` and once it returns the converted string, print the number from `main()` as a string. Note that, you are not allowed to use `'sprintf()'` or `'itoa()'` functions at any part of your program.

Hint: If you are extracting the digits one by one from the integer before storing them as character, it most likely will give you the integer in reversed form. So, you may need to reverse the string before returning the value as string.

Example:

```
Enter the number: 2908
The number as string is: 2908
```

```
Enter the number: -2908
The number as string is: -2908
```

Question-2 (15 points)

Write a C program that asks an user to enter a string `s`. The string should represent an integer, with each character storing one digit of the integer (as the corresponding character). The integer can be positive or negative and can have at max 4 digits. If it is negative, the leftmost (first) character is '-' (the minus sign), and the next character is always a digit from 1-9. If it is positive, it always starts with a digit from 1-9 (*i.e.*, no explicit '+' sign is put). Write a function `'stringToInt()'` that takes the string as input and converts it to the equivalent integer. The string is taken as the input parameter and the return value of the function is the integer. Call this function from `main()` and once it returns the converted integer, print the number from `main()` as an integer.

Note that you are not allowed to use `'strtol()'` or `'atoi()'` function at any part of your program.

Example:

```
Enter the number (as string): 2908
```

```
The number is: 2908
```

```
Enter the number (as string): -2908
```

```
The number is: -2908
```

Question-3 (20 points)

Write a C program that takes a sentence from an user as input and extracts the n^{th} word from it. The user inputs the value of '`n`' (assume the first word means $n=1$). Your program should print the n^{th} word in the console. Please note the followings.

- You can not read a whole sentence that has spaces by `scanf()` function that you generally use. C provides a special `fgets()` function for such purposes. A typical syntax to use it is - `fgets(str, sizeof(str), stdin);` - where `str` is a character array to store the sentence. `str` needs to be declared earlier.
- You can assume a sentence consists of multiple words, spaces, comma (,) and fullstop (.). So, words can be separated by spaces as well as commas. To get the n^{th} word of a sentence, you have to write a function void `get_nth_word(char* str, int n, char* result)`, where `str` is the address of the string that stores the full sentence. `n` is the value of `n` entered by the user and `result` is the address of the character array that will store the n^{th} word after extraction. Naturally, the array `result` will have to be declared in `main` and passed to the function `get_nth_word()` from `main()`. You will have to print the n^{th} word from `main()`.
- Your code should have proper checks so that the user gets proper message if it enters invalid values of `n`.
- You can assume that the sentence can contain maximum 100 characters.

Example:

```
Enter the sentence: Day after tomorrow is Friday, next is weekend.
```

```
Enter the value of n: 9
```

```
Invalid n value.
```

```
Enter the sentence: Day after tomorrow is Friday, next is weekend.
```

```
Enter the value of n: 4
```

```
Word No. 4 of the sentence is: is
```